DATA ENGINEERING · PRESENTATION

From Non-Invasive Diagnosis to Medical Data Engineering

Scaling Breath Analysis with data Engineering tool and making Modern Healthcare Data Platforms

Background: Physics & Computational Medical Research
Present   : Data Engineering Course, DE-FT-July-26
Focus   : Medical data analytics and scalable healthcare pipelines

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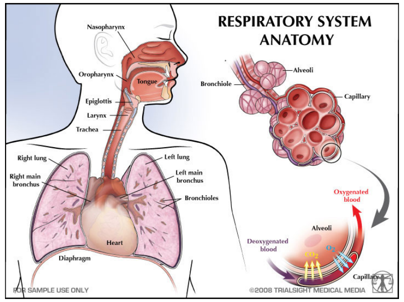
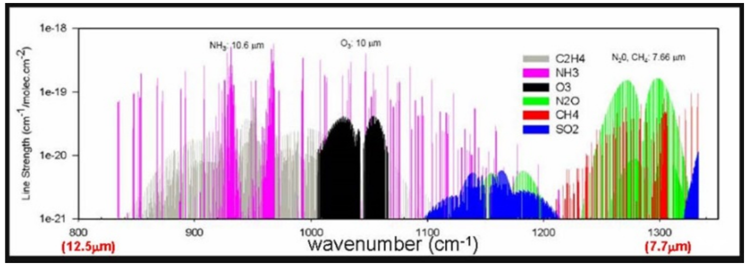
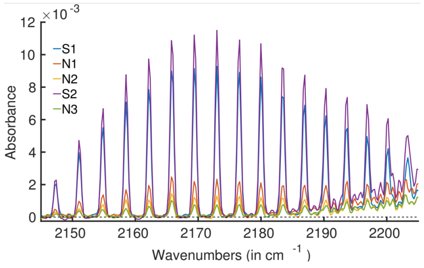
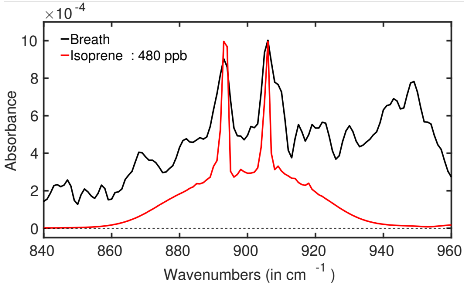
Breath Holds a Diagnostic Fingerprint



Volatile organic compounds in exhaled breath encode metabolic signals linked to disease, captured as high-dimensional spectroscopy data.







How It's a Data Engineering Problem

Spectral curves, clinical metadata, and multi-site cohorts must be ingested, cleaned, versioned, and served — reliably, at scale.

Technology : IR Spectroscopy

Water Suppressed IR Spectra

From Sensor to Clinical Impact:

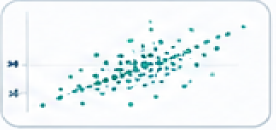
The Real Challenge is Data, Not Just Technology

Turning Breath into Better Health Decisions

THE CHALLENGE

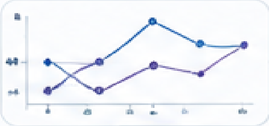
It's not only the sensor or the spectroscopy technique.
It's the **scale, complexity** and **integration** of **real-world data.**

Large Numbers of Measurements
Thousands to millions of breath samples across sites and time.



High-Dimensional Spectral Features
Each sample contains hundreds to thousands of spectral variables.

Repeated Patient Monitoring
Longitudinal data collected over time for the same patient.



Integration with Other Medical Data
Combining breath data with imaging, clinical records, laboratory results and more.

THE SOLUTION: A SCALABLE DATA PIPELINE

A reliable, scalable and reproducible pipeline turns complex data into **trusted** insights.

- **COLLECT** — Multi-source high-volume data
- **PROCESS** — Clean, transform and extract meaningful features
- **STORE** — Secure, scalable and governed data platforms
- **MODEL** — Train, validate and continuously improve models
- **DEPLOY** — Deliver insights for better clinical decisions

- **RELIABLE** — Consistent and accurate
- **SCALABLE** — Handles growth and complexity
- **REPRODUCIBLE** — Transparent and repeatable results
- **IMPACTFUL** — Better diagnosis, better care

KEY MESSAGE: A diagnostic model is useful only if the **data pipeline** is reliable, scalable, and reproducible.

Great models + Great data pipelines = Real clinical impact

Privacy First, Science Always

Building secure, reproducible and FAIR-compliant pipelines without exposing patient data

THE CHALLENGE

Patient data is highly sensitive

Ethics & Consent
Informed consent, ethical approval and data protection are essential.

Privacy & Confidentiality
Patient identifiers and metadata must never be exposed or shared publicly.

Regulatory & Governance
Compliance with GDPR, local laws and institutional policies.

Data Stays Under Control
Preference for on-premise or tightly controlled environments.

Restricted Access
Only authorized users can access data based on roles and permissions.

WHAT DATA ENGINEERING TAUGHT ME

Designing trusted research pipelines

1. Ingest
Collect data from sensors, labs, imaging and clinical systems securely.

2. Validate
Quality checks, de-duplication and completeness validation.

3. Protect
Pseudonymize data, separate identifiers from analytical datasets.

4. Transform & Document
Clean, standardize and transform data with full lineage and documentation.

5. Control Access
Role-based access, audit trails and monitoring.

6. Deliver Reproducible Datasets
High-quality, well-documented datasets for research and model development.

FAIR WITHOUT EXPOSING PATIENTS

FAIR principles in practice

F Findable	Data and metadata are well described and indexed for discovery within controlled environments.
A Accessible	Data is accessible to authorized researchers under defined policies and permissions.
I Interoperable	Use of standard formats and controlled vocabularies enables integration across datasets.
R Reusable	Clear documentation, provenance and quality information ensure data can be reused responsibly.

FAIR does not mean “public to everyone”. It means well-governed, well-described and reusable under appropriate permissions.

KEY MESSAGE

The goal is not to make sensitive medical data public; the goal is to make medical research secure, reproducible and responsibly reusable.

Secure by Design • Governed by Ethics • Powered by Data Engineering

Three Takeaways for Data Engineers

1

Domain data isn't generic data

Functional / spectral data needs custom transforms — not just standard row-and-column ETL. For example, spark can help here

2

Reproducibility is important

A useful health care model needs documented pipeline what can be done from the knowledge learned from Data Engineering course.

3

Privacy and FAIR policy can work together

Sensitive patients information can remain protected, while meta data, standards and access right make research reusable and collaborative

If you are interested few publications link of our research :

1. https://doi.org/10.1016/j.saa.2026.128422//doi.org/10.1016/j.saa.2026.128422, Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy 363 (2026) 128422-----Advancing neonatal care: A non-invasive, contactless method for early diagnosis of neurological disorders

2. https://doi.org/10.1021/acs.analchem.5c04584, Anal. Chem. 2025, 97, 24568−24575-----Is Digital Drying an Effective Method for Infrared Spectroscopy of Gaseous Biofluid?

3. https://doi.org/10.1021/acs.analchem.4c02899 Anal. Chem. 2025, 97, 106−113, Measurement of Bacterial Headspaces by FT-IR Spectroscopy Reveals Distinct Volatile Organic Compound Signatures

4. https://doi.org/10.1038/s41598-021-96845-z , 2021, 11:18381, Towards reliable diagnostics of prostate cancer via breath

5. https://doi.org/10.1038/s41598-019-51417-0 , 2019, 9:161671, Human beings as islands of stability: Monitoring body states using breath profiles

Thank You

Questions & Discussion

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